MQTT Communication Protocol Specification Description(API)

1. Overview of the Agreement

1.1 Objective

This document defines the communication rules between clients and servers based on the MQTT protocol, specifying the connection mechanism, topic format, message flow, and interaction logic, providing a standardized framework for device integration, server development, or system interoperability.

1.2 Scope of Application

This applies to bidirectional data exchange between all MQTT clients (washing machines, dryers, and integrated washing-drying machines) compliant with this protocol and designated servers, covering scenarios such as parameter queries, parameter modifications, viewing and modifying custom programs, device control, and status reporting.

1.3 Core Principles

- Based on the MQTT 3.1.1 protocol standard, it supports QoS levels 0/1/2 (default QoS 1 ensures messages are delivered at least once).
- Supports both SSL and standard TCP connections to meet secure communication requirements across various scenarios.
- The topic uses a hierarchical design, distinguishing message purposes by device ID and function identifier.
- The client and server establish bidirectional subscriptions and targeted communication to ensure precise message delivery.
- Supports reconnection after disconnection (depending on the native features of the MQTT protocol).

2. Basic Definitions

2.1 Term Definitions

term	definition
Client	Refers to machinery such as washing machines, dryers, and integrated washing-drying machines that support WiFi (wireless network) or wired Ethernet ports for communication. Upon startup, they

term	definition
	automatically connect to the server and complete subscription procedures.
Server	Refers to the MQTT server, which requires writing code to intercept messages related to "server," or registering a dedicated client with the fixed name "server" responsible for message reception, processing, and forwarding.
Device ID (Device ID)	The unique identifier for the machine equipment (SN code + machine type code) is assigned by the manufacturer in advance and embedded in the device.
Topic	The routing identifier for messages in the MQTT protocol, specifying the sender and receiver of a message.
Upward Message	The message sent by the client to the server (in Topic format: server/xxx)
Downstream Message	The message sent by the server to the specified client (in Topic format: Device ID/xxx)

2.2 Basic Configuration Requirements

Configuration Options	ask
MQTT Protocol Version	3.1.1
Server Address	Provided by the client (e.g., mqtt://xxx.xxx.xxx.xxx:1883 orssl://xxx.xxx.xxx.xxx:8883)
Server Client Name	Set to "server" (cannot be changed; used by the client to identify the server identity)

Configuration Options	ask
Device ID Requirements	SN Code + Machine Type Code For example: EC16HW00002505100123006-1
Connection Activation	Default: 60 seconds (the client periodically sends heartbeat packets to ensure connection validity)
Authentication Method	Username: Device ID (e.g., EC16HW00002505100123006-1) Password Authentication: Encryption rules must be agreed upon separately with the customer

3. Connection and Subscription Rules

3.1 Client Operation Process

1. Auto-connect upon startup: When powered on, the device automatically initializes and establishes a connection using preconfigured server addresses and authentication credentials. If the connection fails, it attempts up to 60 times; if unsuccessful, it stops trying until the next power cycle or manual connection (the number of attempts can be adjusted via parameters).
2. **Batch subscribe to download data**

3. Connection Reconnection Logic: When the connection is lost (e.g., due to network issues), the client automatically attempts to reconnect

3.2 Server Operation Process

1. Write server-side code: captures Topic messages with the "server" flag, processes them, and forwards them
   perhaps
2. Special Client Registration: Register a special client named "server" to connect to the MQTT server.
3. **Batch subscribe to upload data**
4. Note: If using this mode, configure rules on the MQTT server to prevent other clients from subscribing to messages related to the "server," avoiding accidental message forwarding.
5. Connection persistence: The server must maintain a long-term connection to receive client messages and send commands in real time.

4. Theme and Message Format Specifications

4.1 Theme Classification and Uses

Message Direction	Theme Format	Sender	Recipient	Core Use
Upward	server/runstate	Client	Server	The client reports the operational status (e.g., online status, temperature, machine information, etc.)
Upward	server/param2change	Client	Server	The client reports the parameter modification result (success/failure)
Upward	server/paramshow	Client	Server	Client Reports Parameter Data
Upward	server/control	Client	Server	The client reports the execution result of the reporting command (success/fail

Message Direction	Theme Format	Sender	Recipient	Core Use
				ure).
Upward	server/plan2change	Client	Server	The client reports the program modification results (success/fail ure) in self-report mode
Upward	server/plan2show	Client	Server	The client reports program data in self-service mode.
Upward	server/runrecord	Client	Server	The client reports the execution records of automated programs (e.g., program name, amount, duration, etc.)
Upward	server/getversion	Client	Server	Client reports version number

Message Direction	Theme Format	Sender	Recipient	Core Use
Upward	server/reboot	Client	Server	The client reports the execution result of the reporting command (success/failure)
Downward	DeviceID/param2change	Server	Client	The server sends an instruction to modify the parameters.
Downward	DeviceID/paramshow	Server	Client	The server sends a parameter query instruction.
Downward	DeviceID/control	Server	Client	The server sends device control commands (such as start, stop, adjust duration, upgrade, etc.)
Downward	DeviceID/plan2change	Server	Client	The server sends instructions for

Message Direction	Theme Format	Sender	Recipient	Core Use
				modifying the self-service mode program.
Downward	DeviceID/plan2show	Server	Client	The server sends a program in self-service mode to view the instructions.
Downward	DeviceID/getversion	Server	Client	The server sends a command to check the version number
Downward	DeviceID/reboot	Server	Client	The server sends a command to restart the client.

4.2 Message Payload Format Specifications

- The message body uses JSON format and is transmitted strictly according to the example file format.

5. Key Considerations for Customer Engagement

1. The client has pre-set the device ID and machine type before leaving the factory. If incorrect, contact the manufacturer.
2. Device ID, authentication account, and authentication password must be entered into the server in advance.

3. The server's dedicated client must use "server" as its name; otherwise, it cannot receive uplink messages from clients. Additionally, the server must have proper permissions to prevent other clients from registering or subscribing illegally.
4. All messages must strictly adhere to the JSON format specifications; mandatory fields cannot be omitted, otherwise the message may not be parsed.
5. The server recommends implementing detailed logging functionality.
6. Before connecting, verify the server address, port, and authentication details (username/password, certificate) to ensure correct connection parameters.

Other Attachments

Machine Type Code:

Machine Type	ID code
Single Dryer	0
Hard Mount Washer	1
Soft Mount Washer	2
Stack dryer – Upper dryer	3
Stack dryer – Lower dryer	4
Stack washer dryer – Lower hard mount washer	5

Machine Type	ID code
Stack washer dryer – Lower soft mount washer	6
Combo washer dryer(hard mount)	7
Combo washer dryer（soft mount)	8

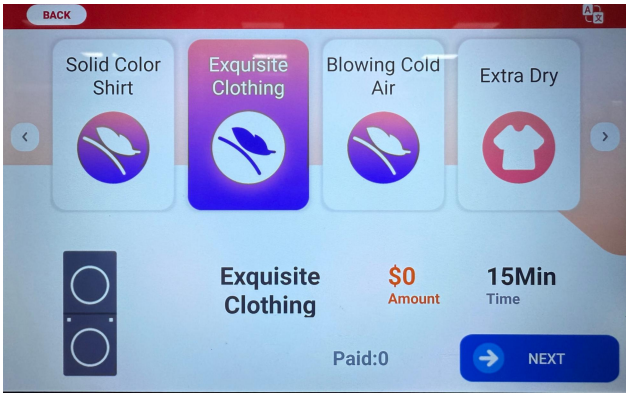
Example of interaction flow data:

Chinese annotated version: mqtt_example_zh.json mqtt_example_zh.json

English annotated version: mqtt_example_en.json mqtt_example_en.json

Touchscreen Function Description





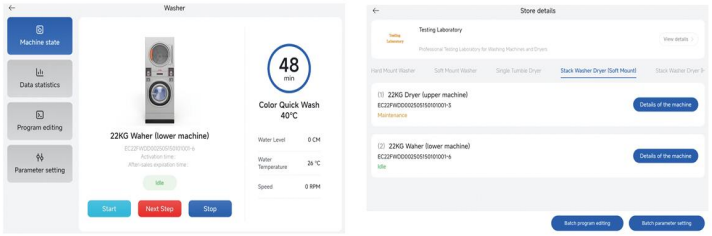
Simple to operate, integrated control



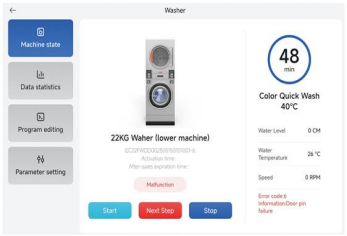
|Energy Saving| Work Efficiency |Stable Performance |

Real-Time Machine Monitoring & Remote Control & Customization

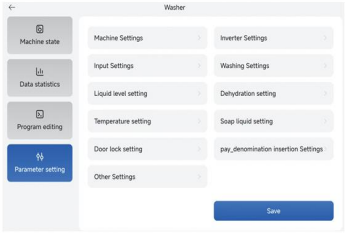
1. Monitor machine statuses (idle, in-use, maintenance) instantly.



2. Receive instant alerts for errors, malfunctions, or low-supply warnings.



3. Adjust machine settings (cycle duration, pricing, modes) remotely.



4. Roll out updates or promotions across all machines with a single click.

